SWAPNIL SHARMA

Ph.D. Candidate in Economics · Boston College

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EDUCATION

Boston College, Boston, MA — *Expected May 2027*
Ph.D. in Economics (Econometrics and Quantitative Economics)

Indira Gandhi Institute of Development Research, Mumbai, India — *July 2019*
M.Phil. in Economics

Indira Gandhi Institute of Development Research, Mumbai, India — *May 2018*
M.Sc. in Economics

Indian Institute of Technology (BHU), Varanasi, India — *May 2012*
B.Tech. in Mechanical Engineering

RESEARCH FIELDS

Macroeconomics; Monetary Economics; Applied Econometrics; Theoretical Industrial Organization

PH.D. WORKING PAPERS

Credit Markets, Uncertainty, and Business Cycles *Job Market Paper* [Link to Paper]

Shows that the shock explaining most business-cycle variation in hours combines two forces with distinct propagation: macro uncertainty causes a broad contraction, while a credit-related shock produces a boom followed by a bust and accounts for the spectral peak in hours. An estimated financial-frictions model, solved with a Deep Equilibrium Network, reproduces both patterns through uncertainty about enforcement and noisy signals about future productivity.

Monetary Policy Surprise and Forward Information

Shows that roughly 10% of high-frequency policy shocks are predictable from forward information held by the Federal Reserve and forecasters. An orthogonalized shock series produces larger and more statistically significant macroeconomic effects than earlier high-frequency studies.

Forward-Looking Credit Pricing and the Endogenous Cycle in Hours Worked *with Sanjay Moorjani* [Link to Paper]

Documents a recurring nine-year cycle in U.S. hours, unemployment, the investment-output ratio, and credit spreads. Shows that the Beaudry, Galizia, and Portier (2020) model generates this cycle through one-period credit pricing based on current employment. Changing only the pricing rule to depend on expected default risk over the loan's life eliminates the endogenous cycle once loan duration extends slightly beyond one quarter. Re-estimation preserves it only up to about 1-1.5 quarters; longer loans require persistent external shocks.

PRE-PH.D. PAPERS

Competition in Two-Sided Markets with Congestion [Link to Paper] [IGIDR Working Paper]

Extends a two-sided platform model to include within-group congestion. Congestion raises platform profits when tolerance is low under single-homing, while the condition reverses under multi-homing.

First Party Content With Sellers Competition in Two-Sided Market [Link to Paper]

Studies how a monopoly platform chooses prices and first-party content when sellers compete. Seller competition can restrict seller participation and reverse standard predictions about first-party-content investment under unfavorable expectations.

Competition among Sellers and Platform Entry into Sellers' Product [Link to Paper]

Compares pure intermediation with a platform that competes in sellers' product market. Entry lowers sellers' membership fees and can benefit all sides under monopoly; with competing platforms, buyer gains depend on product differentiation and perceived quality.

RESEARCH EXPERIENCE

Research Statistician (Consultant), Research Services, Boston College — *Jun 2024–Aug 2026*

- Automated extraction, translation, and structuring of data from 1,000+ scanned Hindi PDFs into analysis-ready datasets.
- Applied machine learning to extract structured, privacy-safe transaction data from 3,000+ heterogeneous credit card statements.
- Consulted on 10+ academic projects involving data acquisition, cleaning, statistical analysis, and modeling.

Research Assistant to Professor Theodore Papageorgiou, Boston College — *Summer 2022*

- Developed and optimized Python algorithms for a high-dimensional fixed-point problem in a macro-labor equilibrium model.

TEACHING EXPERIENCE

Teaching Fellow (Lecturer), Boston College
- Macroeconomic Theory — *Fall 2024*
- Macroeconomic Theory — *Summer I 2025; Summer II 2025*

Teaching Assistant, Boston College
- Principles of Macroeconomics — *Fall 2019*
- International Finance — *Spring 2020; Fall 2023*
- Principles of Economics — *Fall 2022*
- Statistics — *Spring 2022, 2023, and 2024*
- Econometric Methods — *Fall 2026*

Teaching Assistant, IGIDR
- Microeconomic Theory — *Fall 2018*
- Game Theory — *Spring 2019*

ADDITIONAL PROFESSIONAL EXPERIENCE

Intern, Foreign Exchange Department, Reserve Bank of India — *Summer 2017*
- Supported assessment of a $150 million external commercial borrowing for compliance with foreign-exchange regulations.

Assistant Manager, Research and Development, Hero MotoCorp Ltd. — *2012–2014*
- Coordinated development and validation of new motorcycle components for four models.
- Inspected and approved 15+ vendor samples and supported production-capacity expansion.

HONORS AND SCHOLARSHIPS

Boston College, Tuition remission and stipend — *2019–Present*
Indian Institute of Technology Joint Entrance Examination, top 1% — *2008*

TECHNICAL SKILLS

Programming and Data: Python, MATLAB, SQL, Stata, Julia, Microsoft Excel
Agentic AI-assisted research workflows: pipeline development and automation with coding agents, integrated with Python, MATLAB, Stata, and LaTeX
Modeling and computation: DSGE Modeling, Global Solution Methods, Numerical Optimization, Neural Networks, Deep Learning

CERTIFICATIONS

Pricing Options with Mathematical Models, Caltech (Coursera) · *July 2026*
Pandas, Kaggle · *June 2026*

REFERENCES

Rosen Valchev (Chair)
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Peter Ireland
Murray and Monti Professor
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Susanto Basu
Professor
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